

Ember

Status and the path to complete low-mass stellar lifetimes

Scientific and operational handoff for Gregory Laughlin

Physics snapshot: 9 September 2026, 23:09 UTC

Principal result. A fixed-baryonic-mass $0.1M_{\odot}$ model has reached 2.85×10^{12} yr with composition-dependent microphysics and a non-grey gas atmosphere. It remains fully convective and retains $X_{\text{H}} = 0.30289$. This is a reproducible partial hydrogen-burning track, *not* a measured stellar lifetime, a completed blue-dwarf excursion, or a white-dwarf cooling calculation. Here $1\text{Tyr} = 10^{12}$ yr.

The current input-development bottlenecks are atmospheric coverage below $X_{\text{H}} = 0.3$, independently validated hydrogen-poor opacity interpolation, and condensate thermodynamics in layers carrying a small convective flux. The new hydrogen-poor EOS is available separately, but neither it nor the new opacity family is selected by the reference trajectory. Condensate feedback is absent from that trajectory.

Distribution. GitHub carries code, generation specifications, small provenance records and this report. New bulk physics tables, raw histories, source-output archives and checkpoint binaries remain local. The committed generation pipeline is described in [DATA_REPRODUCTION.md](#).

1 The current $0.1M_{\odot}$ model

The initial state is a specified static main-sequence model with $X_{\text{H}} = 0.7$, $X_3 = 0$, $Z = 0.02$, and the GS98 metal pattern. Its age excludes formation and pre-main-sequence contraction. All abundances below use conserved baryonic mass. Values are reported to identify the model; the displayed digits do not represent physical accuracy.

Quantity	Last completed model
Age from the static initial model	2.85Tyr
Mass used in the structure equations	$0.1M_{\odot}$
Radius	$0.148760882R_{\odot}$
Surface luminosity	$0.00185757060L_{\odot}$
Effective temperature	3106.8375 K
Central temperature	6.878544 MK
Central density	$241.23194 \text{ g cm}^{-3}$
Central pressure	$1.6524980 \times 10^{17} \text{ dyn cm}^{-2}$
Surface $\log_{10} g$ (cgs)	5.0930901
X_{H}, X_3, X_4	0.302887598, 0.004663338, 0.672449064
Metal mass fraction	0.02
Convective mass fraction	1 (homogeneous)
Mesh; accepted/rejected macrosteps	512; 1673/1

The minimum midpoint ratio of the gradient required for entirely diffusive transport to the adiabatic gradient is 3.19869. The largest estimated local conductive flux fraction is 0.0328883; the central value is 0.0086111. The maximum reduced ppII/pp reaction ratio is 0.00296634. These are diagnostics of this model, not guarantees about subsequent evolution. The records are the evolution, transport and nuclear [2.85T reports](#).

2 What has been built

Ember is a C++23 reconstruction and extension of the stellar-evolution line recovered in the historical [Heney/F77](#) repository. LBA97 supplies the scientific benchmark and evolutionary picture [1]; it does not supply a lifetime that the new calculation is required to reproduce. The developments below describe Ember’s present implementation, rather than asserting that the older code lacked every underlying physical effect.

Component	Implemented treatment and qualification
Equation of state	FreeEOS 3.0 EOS1 represented by one interpolated Helmholtz potential, with consistent caloric and response derivatives. Molecular formation, ionization, pressure ionization, electron degeneracy and source nonideal terms are included within supported states. The current family includes GS98 metals and explicit H/He3 composition dependence. Potassium, with mass fraction 4.56×10^{-6} , is omitted by this source. Helium isotope corrections retain electronic/cross-section approximations.
Radiative opacity	Composition-dependent AESOPUS 2.1 low-temperature gas opacity and LANL TOPS hot/dense opacity, with supported blends and isotope number-density mapping. The existing interior families cover $Z = .01, .02, .03$; this is a modest interval, not a high-metallicity survey. Substituted source cells and extrapolation are rejected.
Nuclear burning	Explicit non-equilibrium ^3He , reduced ppI/ppII burning, Solar Fusion II S-factor quadrature, finite-degeneracy Salpeter–Van Horn screening, nuclear-neutrino losses and heating derived consistently from atomic mass defects. No pep, hep, ppIII or CNO network; the reduced ppII completion and neutrino approximation are documented.
Transport	MLT with Schwarzschild or full-EOS Ledoux buoyancy. Electron conduction uses Ioffe/Cassisi-family inputs and approximate mixture resistivities. Slow semiconvective and thermohaline composition mixing are available. The semi-convective option has no separate heat flux. No microscopic settling.
Atmosphere	Offline TLUSTY208/SYNSPEC54 LTE, plane-parallel radiative/convective gas models, matched at $\tau = 100$. Molecular and atomic opacity, collision-induced absorption and scattering are included. Runtime interpolation is in H, He3, T_{eff} and gravity. The installed extension has 72 source cells.
Evolution and checks	Analytic zone Jacobians; damped Heney/Newton structure solution; coupled implicit burning and conservative mixing; the discrete thermal first law; adaptive timestep control by step doubling. Fixed mass mesh. Native checkpoints preserve internal state and timestep-controller state; source and executable fingerprints protect reproducibility.

Physical consistency still has identifiable limits. The EOS, atmosphere chemistry, line/continuum opacities and conduction do not all come from one common thermodynamic source. At matching states of the extended gas grid, interior-EOS densities differ from atmospheric-EOS densities by approximately 0.68–1.99%. The common metal inventory improves agreement without removing atomic-weight, partition-function and isotope approximations. Nuclear screening is not derived from the FreeEOS interaction free energy. The conduction mixture uses approximate addition of pure-ion collision effects, and its envelope activation is an explicit domain join rather than a partial-ionization conductivity calculation.

3 The track and the evidence for numerical reliability

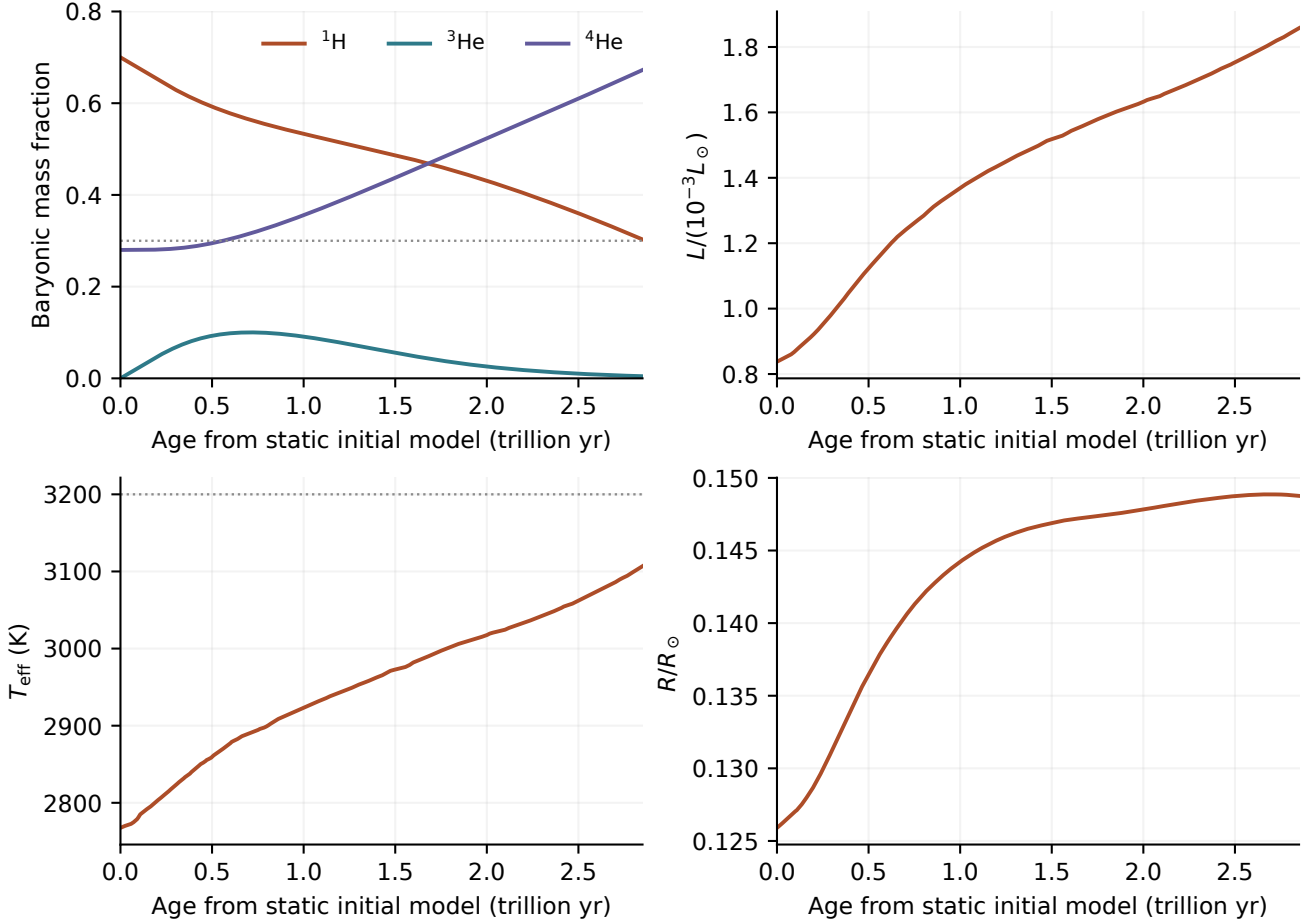


Figure 1: The completed gas-atmosphere trajectory, assembled from the original 0–2.5T run and the 2.5–2.75T and 2.75–2.85T native continuations. The 1674 recorded states include the initial state and 1673 accepted macrosteps. Duplicate restart rows are removed only after equality checks on the overlapping physical states. Dotted lines show the present atmospheric H floor (upper left) and effective-temperature ceiling (lower left). The late, small radius turnover has not been established by late-time numerical refinement.

The star first builds a substantial non-equilibrium ^3He reservoir and later consumes it. A continuation-only maximum must not be reported as the maximum over the entire history. The evolving luminosity, abundance and surface temperature show why a fixed-composition boundary is insufficient for a multi-trillion-year calculation.

Same-physics comparison at 2T	$\Delta L/L$	$\Delta R/R$
512 $\rightarrow$ 1024 mesh points	+0.061632%	−0.012340%
Fourfold tighter time tolerances	−0.003610%	−0.000594%

An independent complete 2T run reproduces the history and profile byte for byte. The checkpoint-enabled executable also reproduces the original 2.5T output exactly. Native restart tests compare subsequent accepted histories and final profiles, including executable relocation and rejection of incompatible inputs. The last 2.75–2.85T continuation took 37 accepted steps and no rejections.

These establish repeatability and numerical sensitivity of the completed segments. They do not establish lifetime convergence, a universal error bound, or physical accuracy of the sources. The energy-balance diagnostics check the implemented discrete first law; they are not an independent proof of an exact global gravitational-energy balance.

4 Input development: accepted, diagnostic, and blocked work

4.1 Hydrogen-poor EOS: available, not selected

A separate GS98 $Z = .02$ family now contains 24 planes: $X_H = .1, .125, .15, .175, .2, .25, .3, .4, .5, .6, .7, .75$ at $X_3 = 0, .12$. All original $X_H \geq .3$ planes are retained byte for byte. A coarser composition spacing produced a 1.22% heat-capacity discrepancy, so extra planes were computed before acceptance. Across 336 fresh source comparisons, maximum discrepancies are 0.04985% in pressure, 0.10866% in energy and 0.19344% in heat capacity. First-law/response checks are below 7.2×10^{-10} . All 1536 sampled states from the 2T, 2.5T and 2.75T profiles give byte-identical old/new EOS probe output. The scope and source manifests are in [FORWARD_EVOLUTION.md](#).

4.2 Hydrogen-poor opacity: under validation

The separate TOPS extension has 30 planes, covering source H fractions .09, .1, .2, .3, .4, .5, .6, .65, .7, .75 at $Z = .01, .02, .03$. The extra .09 plane brackets the atomic/source composition corresponding to baryonic $X_H = .1$ with $X_3 = .12$. Original planes and old-profile evaluations are unchanged; support and derivative checks pass.

An independent midpoint source exposed a real interpolation limitation. At $X = .15, Z = .02$, comparison against the .1/.2 interpolation over 3496 common original source cells, excluding 54 substituted cells, gives a maximum discrepancy of 11.9011%. The worst discrepancy within the audit's defined hot rectangle is 0.918319%. The all-cell maximum is not a measured error in the current star: the active blend and sampled stellar states must be considered. Neither result is a physical uncertainty estimate for TOPS itself.

Four further source fetches completed: $X = .15$ at $Z = .01, .03$ and independent quarter-points $X = .125, .175$ at $Z = .02$. This handoff archives their raw requests and results without importing or accepting a refined runtime family. Rounded filenames `x012/x018` must not be interpreted as actual .12/.18 mixtures. The original midpoint audit must survive any promotion of that midpoint to a new interpolation node.

4.3 Atmosphere and condensates: the current physical block

The installed gas family spans $X_H = .3, .45, .7, X_3 = 0, .12, T_{\text{eff}} = 2600, 2800, 3000, 3200$ K and $\log g = 4.9, 5.15, 5.4$. A separate heldout gas atmosphere gives local interpolation differences of 0.19584% in temperature and 0.03401% in gas pressure at the matching point. The corresponding deeper material-temperature coverage is approximately 1076–7762 K. A proposed extension to 15000 K encountered unsupported native He I 4471 broadening; clipping or extrapolation has not been substituted for valid source coverage.

FastChem equilibrium/rainout diagnostics and coupled gas-depletion experiments have been built. The coupled experiment is an efficient-settling limit with *zero grain opacity*, hybrid depletion/gas chemistry, and no grain enthalpy, size distribution, Mie opacity or time-dependent settling. The last collector

accepted 70 of 72 proposed production cells. Both remaining solar 2800 K, $\log g = 4.9$ corners now have converged solutions but fail the physical grain-enthalpy gate: convective flux fractions in condensing layers are approximately 4.17×10^{-7} ($X_3 = .12$) and 4.11×10^{-7} ($X_3 = 0$), above the unchanged 10^{-8} threshold. Moving to $\log g = 5.0$ did not remove the problem.

Artificial positive heat-capacity perturbations of 10^6 and $10^7 \text{ erg g}^{-1} \text{ K}^{-1}$ at an actual failing corner have completed. They change the matching gas pressure by 7.49×10^{-7} and 7.38×10^{-6} fractionally, respectively. They are useful sensitivity controls, not a full grain EOS or a rigorous bound on missing physics. There is no installed condensate runtime family or install receipt.

5 Physical requirements along the remaining evolution

The expected isolated-star path is prolonged helium enrichment, a hotter excursion, a radiative/conductive core, central H exhaustion, residual shell burning and a helium-remnant cooling sequence [1, 2]. Each stage changes the applicability of current approximations.

5.1 Radiative-core formation and redistribution of fuel

Once full convection ends, microscopic gravitational settling and thermal/chemical diffusion can compete with burning and boundary motion. Their timescales need evaluation over the actual density, temperature and composition profiles. Resolving moving convection boundaries, steep composition gradients and thin shells may require adaptive remeshing. Semiconvective heat transport and the sensitivity to mixing efficiencies also need attention. These are acknowledged roadmap tasks; the implemented Ledoux/mixing tests do not yet demonstrate the evolved stellar transition.

5.2 Exhaustion and residual shell burning

All input families eventually need support approaching zero H, including nearly hydrogen-free core material adjacent to a fuel-bearing shell. The reduced ppII completion assumption needs reassessment when protons become scarce. The missing pep and other pp branches, CNO catalysts at higher temperature or metallicity, and screening/EOS consistency require quantitative error assessment. The nuclear module currently limits its S-factor expansion to $T \leq 20 \text{ MK}$ and rejects quantum-dominated screening states.

Central H exhaustion, cessation of sustained H burning and a later cooling luminosity must be separate measured endpoints. A residual envelope can remain energetically relevant after central exhaustion. Diffusion alters residual burning in helium-white-dwarf calculations [3]; some binary-formed objects experience unstable CNO shell flashes [4]. These identify mechanisms to test, not envelope masses or flash thresholds to transplant into an isolated $0.1M_{\odot}$ star. A flash is not presumed for this reference case.

5.3 The atmosphere of the emerging remnant

A helium-rich interior does not determine the photospheric composition. Residual hydrogen can float upward; later convection can dilute or mix it into helium-rich material. White-dwarf transport calculations demonstrate both processes [5]. A future boundary treatment must therefore follow the resolved surface composition and envelope history.

Cooling brings the object back into cold-atmosphere regimes at different gravity and density. Relevant inputs include nonideal H/He chemistry, pressure-distorted collision-induced absorption, dense-fluid continuum opacity, line broadening and possibly refraction [6]. The current dwarf atmosphere grid and its condensate experiment do not establish this coverage. Remnant-envelope transport and dense cool atmospheres lack an operational implementation plan beyond the broader source-extension goals.

5.4 Cold ion thermodynamics and remnant cooling

Electron degeneracy and thermal/gravothermal evolution already exist in Ember. Further cooling requires appropriate strongly coupled ion thermodynamics, crystallization and latent heat where applicable, mixture separation, ionic quantum corrections to heat capacity, and conductivity through the relevant liquid, solid and partially ionized regimes. Quantum effects can modify the liquid heat capacity before freezing [7].

A helium remnant requires its own density- and composition-appropriate phase physics. Carbon–oxygen phase separation or a standard ^{22}Ne sedimentation prescription must not be assigned automatically to a star that never underwent helium burning. Dense-ion and solid terms are anticipated by the architecture, but those hooks are not implemented cooling physics.

Full thermal-neutrino losses remain absent from the stellar energy equation. At 2.85T a plasmon-only post-processing estimate gives $L_{\nu,\text{plasma}}/L = 1.45 \times 10^{-8}$; it does not evaluate all thermal channels or establish insignificance throughout later contraction. Their importance must be checked rather than inferred from present-day conditions.

6 Effects outside the immediate grid programme

6.1 Mass, rotation and magnetic history

The reference star has fixed baryonic mass and no wind, angular-momentum, magnetic-convection or starspot evolution. Even weak time-averaged mass loss can accumulate over trillions of years. For scale only,

$$\langle |\dot{M}| \rangle = 10^{-15} M_{\odot} \text{ yr}^{-1}, \quad \Delta t = 5 \times 10^{12} \text{ yr} \implies \Delta M = 0.005 M_{\odot}.$$

This is 5% of the reference mass; it is not a prediction that such a wind persists. The relevant task is to bound the integrated loss under plausible rotation and magnetic histories, rather than extrapolate a present-day rate unchanged. These are presently controlled omissions without implemented models.

The integrated nuclear rest-mass release by 2.85T is 0.00282709 of baryonic mass. Heating accounts for that release, but Newtonian gravity still uses conserved baryonic mass. This approximation needs assessment before a precise cross-code lifetime comparison; mesh refinement cannot remove it.

6.2 A mass–composition survey includes different branches

The initial driver, detailed EOS and atmosphere currently target one mass and $Z = .02$. Helium enrichment by H burning at fixed Z is distinct from a grid of stars born at high Z . The fundamental survey is in mass, initial H/He abundances and metal pattern. A $\text{lifetime}(M, Z)$ curve selects a relation $Y(Z)$ and relative metal abundances. Independent high- Z EOS/opacity validation is essential; source availability is not proof of accuracy.

Near the minimum hydrogen-burning mass, initial entropy and contraction must be tested to distinguish a stable star from a cooling substellar object. At larger masses, early radiative regions, shell-driven giant expansion and, if the chosen mass/composition permits, helium ignition require additional coverage and physics. Repeated ${}^3\text{He}$ -driven convective episodes near the fully convective boundary are a further transport-resolution test [8]. They are not missing ${}^3\text{He}$ chemistry in the present network. No giant phase, helium flash or carbon–oxygen remnant is presumed for the isolated $0.1M_{\odot}$ case.

The metallicity-dependent lifetime maximum discussed by Adams and Laughlin [2] is a hypothesis to bracket numerically. Its location and even the relevant survey cut depend on helium enrichment, metal pattern and mass. Neither a specific maximum nor the historical roadmap’s nominal 12T duration is imposed as an answer.

6.3 The much more remote future

Encounters, mergers, accretion and the evolving radiation environment require an environmental model. Possible dark-matter heating requires explicit particle and halo assumptions. Cold pycnonuclear processing lies outside the current thermal network, which switches burning off below 10^5 K. Hypothetical proton decay adds a different energy, mass and composition evolution problem. Such conditional extensions belong to a later future-universe programme [2]; none is required to claim an isolated-star hydrogen-burning lifetime once its stated physical assumptions have been tested.

7 Priorities and completion criteria

1. **Preserve the reference and finish source validation.** Retain the accepted gas inputs and failed controls. Compare the completed TOPS refinement sources on the relevant opacity/blend domain with independent compositions; publish a separate validated family. Resolve grain enthalpy/convection or justify a physically controlled alternative before altering production policy.
2. **Provide the next atmospheric coverage.** Extend H, temperature and gravity along the star’s actual path; independently validate interpolation and source support. Add explicit input selection. Avoid repeatedly calculating irrelevant corners solely to fill a large rectangular grid.
3. **Follow the structural transition.** Compare diffusion, nuclear and transport timescales; implement significant missing physics. Resolve core and shell boundaries and demonstrate spatial/timestep convergence there. Preserve conservative abundance and energy accounting.
4. **Measure and converge lifetimes.** Record central exhaustion and the end of sustained burning separately. Vary endpoint thresholds, mesh, timestep, source-grid resolution and initial models. Repeat/restart and compare independent codes with matched mass, abundance and age conventions.
5. **Generalize mass and composition.** Bracket any lifetime maximum with physically validated high- Z source families; assess $Y(Z)$, metal-pattern, mass-loss and mixing sensitivities.
6. **Develop the remnant programme.** Couple surface composition evolution to dense atmospheres, conduction and ion thermodynamics. Validate cooling milestones before undertaking speculative environmental or particle physics branches.

8 Compute, reproducibility and operational handoff

The C++ Release build uses native arm64, `-mcpu=apple-m4` where supported, `-O3` and floating-point contraction, without `-ffast-math`. Accelerate is linked; the small Henyey blocks currently use a portable pivoted kernel. There is no GPU implementation. Atmosphere parallelism is across independent serial source jobs; opacity generation was capped separately. The external Fortran atmosphere build is arm64 but the recorded source benchmark does not use an M4-specific compiler target.

The expensive radiative-transfer/chemistry models are computed offline and interpolated during stellar evolution. One isolated near-converged atmosphere replay used 86.56 CPU seconds: formal solutions 38.69, radiative matrix 16.86, energy/convection 27.64, and dense solve 0.00334 seconds. Chemical-kernel work accounted for 46.79 seconds *within* those phases (54%); it must not be added again. This is one sampled replay, not a grid-wide performance estimate. An earlier one-trillion-year stellar run took about 10.2 minutes on the M4 Max. The recorded evidence is [nongrey_cpu_profile.json](#).

Local recovery artifacts. The committed `recovery_manifest.json` records compressed and uncompressed SHA-256 hashes for local copies of the three complete evolution segments, their receipts, the final native checkpoint and the exact original macOS arm64 executable. The corresponding ignored `artifacts/` directory is not included in a fresh GitHub clone. The final raw evolution JSON hash is

```
b2d67c169fb29d6a87b2fb66ea8722427a2ce35c8f6b9dc475a6b2b6f15ed0b4.
```

The original executable hash is

```
d04ae649221f0d35c448e438fea620cb72089c6a983498b34c0b2fad7877f974.
```

The figure is reproducible directly from the committed compact CSV with `build_figures.py`. Its `--from-archives` option rechecks local raw-segment joins and rebuilds the CSV. Physics tables require the generation pipeline or a separately supplied local archive. Source licenses and provenance remain unchanged; previously published data history is retained.

Restart contract. Native restart requires identical physical selections, tolerances, mesh, executable bytes and original table bytes. Changing physics requires a new calculation under the current implementation; the old state cannot be silently reinterpreted. A fresh build is not presumed byte-identical. Keep the original binary and use separate development builds and output paths. The present H floor leaves little further evolution under unchanged inputs. See [RESTART.md](#).

Jobs at handoff. At the 23:09 UTC inspection, logs show the TOPS raw refinement fetch and both artificial heat-capacity controls completed. The last solar low-gravity atmosphere failed the grain-enthalpy audit after convergence. The old installer was stopped, and no condensate install receipt existed. A queued stellar controller may still be waiting for that receipt. These log observations do not certify current process liveness. No controller was stopped or restarted in preparing this report; inspect before starting new work.

Verification and next session. A separate Release build succeeded; all 28 CTest suites and 13 additional condensate-import cases passed using the installed local tables. A fresh clone requires those inputs for table-dependent checks. Results are recorded in the adjacent `validation.json`; the original checkpoint executable is preserved. The report is compiled with Tectonic and checked for missing references and layout problems.

The copyable next-session instruction is [NEXT_SESSION_PROMPT.md](#); the latest operational checkpoint is at the beginning of [HANDOFF.md](#). Earlier checkpoint entries are historical. The report and source manifest are a dated snapshot; later source changes must receive their own receipts and validation.

References

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- [8] van Saders, J. L., & Pinsonneault, M. H. (2012). A ^3He -driven instability near the fully convective boundary. [arXiv:1201.5381](#).